

Sanjith S J

+91 90877 30234 | sanjith@bettim.tech | github.com/BETTIM | leetcode.com/BETTIM

Education

Amrita Vishwa Vidyapeetham

B.Tech in Computer Science **CGPA: 8.2/10**

Coimbatore, TN

Jul 2024 – May 2028

- **Relevant Coursework:** Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, DBMS, Computer Networks

Experience

Titan Company Ltd

Apr 2026 – May 2026

Software Development Engineer Intern

- Built multimodal jewelry image analysis pipeline (Google Gemini, CLIP vision, OpenCV) extracting **14+ structured attributes** per image (type, stone counts, metal, setting) with confidence flags for internal evaluation workflows.
- Developed **4 Next.js applications** for jewelry pricing (live bullion MRP with wastage/making/GST, Tanishq PDP scraping + RAG in Supabase, design-to-render) and engineered GenAI pipeline with **3 parallel prompt agents**, parent-judge selection, **7-model** Imagen fallback, and NDJSON streaming REST API.
- Trained retail machine learning models on **73K+ store-product records** achieving **R² 0.99** demand forecasting (RandomForest) and **97% ROC-AUC** stockout classification (GradientBoosting), with KMeans segmentation and IsolationForest anomaly detection.

Amrita School of Computing

Sep 2025 – Present

Undergraduate Teaching Assistant

Coimbatore, TN

- Designed and deployed a Python-based autograding system using PyTest, reducing grading turnaround from **~2 weeks to <1 hour** for **500+ students**.
- Built a scalable full-stack TA management portal using Next.js and Supabase, streamlining onboarding and evaluation for **50+ candidates/semester**.
- Analyzed Git commit histories across **15+ teams** to detect collaboration anti-patterns and improve evaluation strategies.
- Engineered automated reporting pipelines improving grading consistency and eliminating manual data-entry errors.

Projects

SaaS Event Synchronization Engine | Python, Next.js, PostgreSQL, AWS Lambda, OAuth 2.0

2025

- Architected an event-driven platform synchronizing schedules between WhatsApp, Google Calendar, and Notion in real-time.
- Built fault-tolerant ingestion pipelines using AWS Lambda and webhooks to process asynchronous events with **99.9% reliability**.
- Implemented secure OAuth 2.0 authentication and token management across multiple third-party REST APIs.

High-Performance Skip List Engine | Java, Spring Boot, Maven, D3.js

2025

- Implemented a custom Skip List achieving $O(\log n)$ average-time operations for search and insertion.
- Developed an interactive D3.js visualization demonstrating probabilistic balancing on datasets exceeding **10k nodes**.
- Exposed RESTful APIs via Spring Boot to dynamically transform linear data structures into optimized Skip Lists.

Titan Inventory Engine | FastAPI, PostgreSQL, React, Docker

2026

- Architected full-stack inventory planner for a **10-store × 100-SKU** matrix with seeded demand, lead-time, and vendor data in PostgreSQL.
- Implemented two-stage optimizer computing ROP/safety stock/EOQ plus demand-weighted floor-area allocation, validated via weekly Monte Carlo stockout simulation.
- Delivered React dashboard with store digital-twin views, Recharts analytics, and **15+ REST endpoints** via FastAPI; containerized with Docker Compose.

Technical Skills

Languages: Java, Python, TypeScript, C++, SQL

AI/ML: Google Gemini, Imagen, scikit-learn, OpenCV, Computer Vision, RAG/embeddings

Backend: FastAPI, Flask, Spring Boot, REST APIs, OAuth 2.0, AWS Lambda

Frontend: React, Next.js, Tailwind CSS, D3.js, Recharts, Plotly

Data & Infra: PostgreSQL, Supabase, Docker, Git, AWS, GCP, Linux